

Completion Report: oic-stock-alert

Date: 2026-05-25 | Level: Starter | Format target: PDF blueprint

Repository: <https://github.com/ai-ml-kiosk/test-oic-stock-api>

1. Summary

1.1 Feature Overview

The OIC Stock Alert MVP is a dependency-free Python HTTP API that evaluates stock alert rules and returns OIC-friendly JSON routing metadata. It provides a health endpoint, an alert evaluation endpoint, deterministic mock quote behavior, validation, rule evaluation, error mapping, OpenAPI/JSON Schema artifacts, sample payloads, and an OIC Switch branch mapping guide.

The API is designed for Oracle Integration Cloud flows that need a stable branch field:

```
response.oic.switchBranch
```

Supported branch values:

- ALERT_TRIGGERED
- NO_ALERT
- INPUT_ERROR
- QUOTE_UNAVAILABLE
- PROVIDER_TIMEOUT
- SYSTEM_ERROR

1.2 Final Match Rate

95% (Target: 90%)

The Check phase found 38 implemented or substantially matched items out of 40 design items. The remaining gaps are accepted MVP limitations related to live quote provider selection and live-provider timeout behavior.

2. Related Documents

Document | Path

Plan | docs/01-plan/features/oic-stock-alert.plan.md

Design | docs/02-design/features/oic-stock-alert.design.md

Do notes | docs/02-design/features/oic-stock-alert.do.md

Gap analysis | docs/03-analysis/oic-stock-alert.analysis.md

Report markdown | docs/04-report/oic-stock-alert.report.md

Report PDF | blueprints/oic-stock-alert-completion-report.pdf

3. Completed Items

- [x] Planned MVP scope, functional requirements, non-functional requirements, success criteria, and risks.
- [x] Designed API architecture, data model, JSON payloads, error contract, mapping logic, and OIC Switch branches.
- [x] Implemented GET /health.

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- [x] Implemented POST /v1/alerts/evaluate.
- [x] Implemented request normalization for request ID, symbol, currency, provider mode, and OIC tracking ID.
- [x] Implemented validation for symbol format, rule count, duplicate rule IDs, metrics, operators, thresholds, severity, message length, and JSON body size.
- [x] Implemented deterministic mock quote provider with success, missing quote, timeout, and provider error fixtures.
- [x] Implemented alert evaluation for gt, gte, lt, lte, eq, and absGte.
- [x] Implemented aggregate decisions for ALERT_TRIGGERED and NO_ALERT.
- [x] Implemented error decisions for INPUT_ERROR, QUOTE_UNAVAILABLE, PROVIDER_TIMEOUT, and SYSTEM_ERROR.
- [x] Implemented OIC metadata: switchBranch, notificationRecommended, retryRecommended, and trackingId.
- [x] Extracted OpenAPI 3.1, JSON Schemas, sample payloads, and OIC Switch mapping artifacts under artifacts/.
- [x] Added unit, API contract, and artifact tests.
- [x] Published implementation and artifacts to GitHub.

4. Deviations From Design

Deviation | Impact | Decision

Live quote provider integration is not implemented. | Low for MVP; mock mode is the approved local/CI path. | Defer until provider is selected.

QUOTE_PROVIDER_TIMEOUT_MS is loaded but not used for outbound calls. | Low; timeout is simulated through mock symbol TIMEOUT. | Implement with live provider.

QUOTE_PROVIDER_NAME is loaded but not directly surfaced in diagnostics. | Low; responses include provider mode and quote provider identity. | Revisit when live providers are added.

PROVIDER_ERROR retry behavior is conservative (false). | Low; avoids unsafe retry loops without provider-specific classification. | Add provider-specific retry mapping later.

Extracted artifacts were added after design. | Positive addition. | Keep artifacts as blueprint deliverables.

5. Metrics

Metric | Value

Final match rate | 95%

Target match rate | 90%

PDCA iterations | 1

Tracked files | 31

Python source lines | 660

Test source lines | 245

Artifact lines | 1,256

Documentation lines | 995 before this report

Automated tests | 16

Test command | python3 -m unittest discover -s tests

Latest pushed commit | ab692f4 Add OIC stock alert gap analysis

6. Quality Metrics

Area | Result

Unit tests | Passing

API contract tests | Passing

Artifact tests | Passing

JSON artifact parse checks | Passing

Secrets committed | None identified

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Runtime dependencies | None

OIC Switch branch coverage | Complete for all designed branches

7. Blueprint Artifacts

Artifact | Path

OpenAPI contract | artifacts/openapi/oic-stock-alert.openapi.json

Request JSON Schema | artifacts/json-schema/alert-evaluation-request.schema.json

Response JSON Schema | artifacts/json-schema/alert-evaluation-response.schema.json

Sample alert request | artifacts/samples/evaluate-alert-triggered.request.json

Sample alert response | artifacts/samples/evaluate-alert-triggered.response.json

Sample no-alert request | artifacts/samples/evaluate-no-alert.request.json

Sample no-alert response | artifacts/samples/evaluate-no-alert.response.json

Validation error sample | artifacts/samples/validation-error.response.json

Provider timeout sample | artifacts/samples/provider-timeout.response.json

OIC Switch guide | artifacts/oic/oic-switch-branches.md

8. Lessons Learned

1. Keeping the MVP dependency-free made local testing and GitHub publication straightforward.
2. The OIC Switch branch field should remain a first-class contract field because it simplifies integration mapping.
3. Extracting OpenAPI, JSON Schema, and sample payload artifacts made the blueprint more reusable than a design document alone.
4. Mock provider fixtures are enough to validate OIC routing paths before selecting a live market data provider.
5. Live-provider retry semantics should be provider-specific rather than guessed early.

9. Follow-up Items

- [] Select the live stock quote provider.
- [] Implement live quote provider integration.
- [] Apply QUOTE_PROVIDER_TIMEOUT_MS to live outbound calls.
- [] Surface provider name in diagnostics when multiple providers are supported.
- [] Add provider-specific retry classification for transient errors.
- [] Decide whether OIC or the API owns persistent watchlists in a future release.

10. Final Recommendation

Proceed to archive after stakeholder review of the PDF blueprint. The MVP meets the PDCA report threshold and is ready for reuse as an OIC Stock Alert API blueprint.